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ArthurBernard

Arthur Bernard

Data Scientist in Quantitative Finance



Key skills

- Proficiency in **statistics & econometrics softwares**: R (Advanced), Octave/Matlab (Advanced), SAS (Beginner), STATA (Beginner).
- Operating systems: **Unix** and **Windows**.
- Languages: French (**native speaker**), English (**professional level**).
- Database: Basic knowledge of **SQL** and **NoSQL** databases.

Programming

- Highly advanced: **Python** (expertise in NumPy, Pandas, Cython, PyTorch, Keras, Sickit-Learn, Asyncio, Multi-process/thread, etc).
- Advanced: **Shell**, **VBA**, $\text{\LaTeX}$.
- In progress: **C++**.

MOOCs

- Learn to program with **Python**, on OpenClassRooms.
- **Machine Learning**, by Andrew Ng, on Coursera.
- **Deep Learning**, by Andrew Ng, on Coursera.
- And other diverse courses about **Linux**, **C++**, etc.

Interests

- Artificial intelligence.
- Crypto-currencies/Blockchains.

Experiences

- Jan. 2019 – **Quant Researcher** *at Napoleon Group*, Paris
Oct. 2019 R&D of trading strategies, **multivariate prediction** with neural networks, **execution order algorithms**, development of **backtesting** and **financial analysis** tools, and webscraping data.
- Jun. 2018 – **Intern in Quantitative Finance** *at Napoleon Group*, Paris
Dec. 2018 Research of quantitative strategies and **portfolio allocation** algorithms. **Data-science competition** elaborated for the Collège de France.

Personal projects

- 2018 – 2019 **Machine/deep learning tools adapted to finance**
Development of a Python and Cython package to create **neural networks**, **backtest strategies**, analysis with **econometric models** and **financial indicators**, etc. Published on PyPI as *fynance*.
- 2017 – 2018 **Webscraping package**
Development of a python package to **download data** and **update database** from some crypto-currency exchanges. Published on PyPI as *dccd*.
- 2016 – 2019 **Trading bot algorithms on crypto-currencies**
Development and maintenance of trading bots with Python and Bash scripts. Starting in 2016 with **arbitrage strategy**, and more recently create **strategies with neural network**. Partly available on my GitHub in the repository *Trading_Bot*.

Education

- 2017 – 2018 Master's Degree **Econometrics of Banking and Financial markets** *at Aix-Marseille School of Economics*, Marseille
Courses: Stochastic finance, financial econometrics, financial engineering, econometrics of exchange rates, neural network, etc.
Projects: Intraday analysis of BTCUSD versus EURUSD, etc.
Master thesis: Analysis of dynamics of Bitcoin.
- 2013 – 2016 Bachelor's degree **Economics and Management** *at Aix-Marseille University*, Marseille
Specialization: Finance.
Courses: Time series econometrics, financial markets, statistics, optimization, informatic (SQL and VBA), etc.
- 2012 A-Level **Science** *at High-School M. M. Fourcade*, Gardanne

Miscellaneous

- 2019 **Data-science competition** *at ENS Challenge Data*
6th out of more than 100 competitors, about prediction of daily stock movements on the US market, proposed by **Capital Fund Management**.
- 2013 – 2018 **Director** *at Mutuelle des Etudiants de Provence*, Marseille
Approval of budgets, financial investments, internal policy, etc.
- 2014 – 2016 **Founder and general secretary of student association**
Organisation and management of team projects.
- Present **Hobbies**
Cooking, travelling (Norway, Scotland, Eastern countries, etc.), swimming (competition) and theater.

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